Curriculum Vitae

Professor Karen Knapp BSc (Hons), PCAP, PgC, PhD, PFHEA
Department of Health and Care Professions
Faculty of Health and Life Sciences
h-index (Google Scholar) 33, i10-index 67

Academic Qualifications

- BSc (Hons) Diagnostic Radiography (1995)
- PhD Radiological Sciences (2002)
- PgC Academic Practice (2005)
- PgC DXA Reporting (2014)
- ASPIRE Senior Fellowship and SFHEA (2012)
- ASPIRE Principal Fellowship and PFHEA (2024)

Professional Registration

- HCPC –Radiographer (D) RA35166
- Member of the Society and College of Radiographers 46300

Current Position

- Professor of Musculoskeletal Imaging (2024), University of Exeter (Jul 2024-)
- Head of Health and Care Professions, University of Exeter (Aug 2022-)
- Bank DXA reporting Radiographer, Dorset County Hospital (Feb 2025-)

Previous Positions

- Associate Professor in Musculoskeletal Imaging (2014-2024), University of Exeter
- DXA Reporting Radiographer (ad-hoc), InHealth (2021-2024)
- DXA reporting Radiographer (bank) Royal Devon and Exeter Hospital (2019-2020)
- Head of Medical Imaging (2013-2022), University of Exeter
- Senior Lecturer (2006-2014), University of Exeter
- Director of Education for Medical Imaging (2010-2013), University of Exeter
- Head of Radiography Education (2007-2010), University of Exeter
- Programme Co-Lead DClinRes (2009-2014) , University of Exeter
- Programme Lead for BSc (Hons) Medical Imaging (Diagnostic Radiography (2004-2007), University of Exeter
- Lecturer in Medical Imaging (2004-2006), University of Exeter
- Post-doctoral research fellow King's College London (2001-2004)
- PhD Student (1997-2002)
- Nuclear Medicine Radiographer, Guy's & St Thomas' NHS Trust (1996-1997)
- Bank Radiographer, Medway NHS Trust (1996-2001)
- Nuclear Medicine Radiographer, Medway NHS Trust (1995-1996)

Honorary Contracts

- Jul 2018 – Feb 2023 Visiting Professor at the University of Suffolk (stepped down as commitments had completed)
- Oct 2018 – present Honorary radiographer, Nobles Hospital, Isle of Man
- Jun 2006 – present Royal Devon University Hospitals Foundation NHS Trust (Research)
- May 2014- May 2015 Honorary Radiographer in Bone Densitometry, Plymouth Hospitals NHS Trust

- May 2007 – Jan 2012 South Devon Healthcare NHS Trust (Torbay Hospital) (Research)
- April 2009 – April 2010 Honorary specialist advisor in Bone Densitometry, Plymouth Hospitals NHS trust
- Oct 2006 – Oct 2008 Guy's and St Thomas' NHS Trust (Research)

Research and Publications

My research covers a broad range of areas, but I am best known as an expert in osteoporosis and diagnostic techniques relating to this area.

Grants for the last 10 years

1. **2026-2027**. Co-I **Worldwide Universities Network (WUN)**. AI for Well-Being: Evaluating Synthetic Data from Menopause Narratives. £30,427.49

2. **2025-2027.** Co-I **European Space Agency (via Adaptix Ltd)** 3D Bone Quality Imaging for Space. £80,000

3. **2025-2029 NIHR HTA** Co-I Randomised controlled trial of the clinical and cost effectiveness of small bore, Seldinger, versus large bore, surgical chest drains for the treatment of traumatic haemo/pneumothoraces in the Emergency Department £2,933,733.02

4. 2023-2026. Co-I **NIHR INSIGHT**: Inspiring Students into Research (R1 2023). £3,035,525

5. 2023-2025: **Innovate UK –Advancing Precision Medicine**. UoE PI Earlier identification and enhanced characterisation/quantification of Osteoporosis and Rheumatoid Arthritis for Clinics/MIU/ED/CDCs ["DT-RHU"] £996,967

6. 2023-2026 **Gilead** - Co-I. Evaluating the efficacy of REDACTED for the improvement of lung function, perfusion, and symptom profile in long-covid patients. £1.25M

7. 2023-2024 **MRC** Co-I Assessing the potential to transform our 'Just A Minute' of bone specific activity finding into a population level intervention to help preserve bone £62,037

8. 2022-2025 **Animal Free Research UK**.Co-I An evaluation of naturally-occurring plant-based sex hormone mimetics as senotherapeutic drug candidates. £180,000

9. 2022-2024 **Royal Osteoporosis Society**. Co-I How can understanding of DXA scans among primary care practitioners & patients be optimised to potentially increase treatment uptake? £62,627

10. 2022-2022 **Royal Osteoporosis Society**. Co-I Assessing the validity and acceptability of using smartphones to monitor bone-specific activity (BSA) £10,000

11. 2020 **Wellcom**e **ISSF2** R4 Co-I Early identification of haemochromatosis using deep learning £3,410 (Internal)

12. 2019 **Wellcome / EPSRC ISSF2** Seedcorn R3 PI Computer Surveillance Program in Neck Injury Evaluation: The CSPINE Study Feb 2019 to July 2019 £9,300 (Internal)

13. 2018-2019 **Royal College of Emergency Medicine** Co-I Ultrasound Directed Reduction of Colles Type distal radius fractures in ED (UDiReCT) RCEM Grant Number: G/2018/5 £9,993

14. 2017-18 **RD&E Small grants fund** Co-I Prediction of Failure of Emergency Department Closed Manipulation of Colles' Distal Radial Fractures (PREDICT) £4,604

15. 2018-2023 **Stryker** Co-I A randomized controlled trial into un-cemented versus cemented knee arthroplasty. £36,000

16. 2017 **Science – Translational Medicine Theme** Co-I Implementing a novel accelerometer-based method to quantify the association between habitual physical activity and direct measures of bone health. £10,000 (Internal)

17. 2017 **Innovation, Impact and Business Collaboration Grant**, University of Exeter Co-I With additional match funding from DM Orthotics, the NIHR funded Dignity Healthcare Technology Co-operative (D4D) Undertaking PPI to develop spinal orthoses for patients with osteoporosis-related vertebral fractures. Jan-Jul 2018 £10,000 (Internal)

18. 2018 **Wellcome ISSF2 Seedcorn R3 Co-I** Citizen Segmentation: Can non-experts help train computers to segment medical images? £9,902 (Internal)

19. 2016-2025 **Stryker** Co-I RCT analysing effects of fixation strategies and effect on loading and bone stock £231,000

20. 2016-2018 **College of Radiographers** PI Enhancing the professional practice of radiography practitioners using a formal research mentorship scheme: a pilot action research study Jan 2016 to March 2018 £14,950

21. 2015 **RD&E Small grants fund** Co-I The CSPINE study: investigating the acceptability and efficacy of computer-aided detection software for cervical spine fracture diagnosis. £10,201

22. 2014-2015 **EPSRC** Co-I Computer-Aided Detection of Cervical Spine Injuries: A Feasibility Project March 2014-March 2015 £124,044

Publications

Books (edited)

1. Shayne Chau, Emma Hyde, **Karen Knapp**, Christopher Hayre. Person-Centered Care in Radiology. Routledge Taylor and Francis. In print – released 2024

2. Carver E, Carver B and **Knapp KM**. Carvers' Medical Imaging. Third Edition. Elsevier 2021 ISBN 978-0-7020-6955-0

3. Maryann Hardy, **Karen M Knapp**, Aarthi Ramlaul, Christina Malamateniou. Artificial Intelligence for Radiographers: Basic Principles, Clinical Applications and Implementation Considerations. Publication date: 2026/1/15 Springer Nature Switzerland

Chapters in books

1. Christina Malamateniou, Maryann Hardy, Karen M Knapp, Aarthi Ramlaul. Radiographer Professional Body Contributions. In Artificial Intelligence for Radiographers: Basic Principles, Clinical Applications and Implementation Considerations. Publication date: 2026/1/15 pages 275-289 Springer Nature Switzerland

2. Maryann Hardy, Karen M Knapp, Aarthi Ramlaul, Christina Malamateniou. Preparing for a Future with AI. In Artificial Intelligence for Radiographers: Basic Principles, Clinical Applications and Implementation Considerations. Publication date: 2026/1/15 pages 291-301 Springer Nature Switzerland

3. Chapter Eight – Person Centred care in DXA and Secondary Fracture Prevention Jill Griffin and **Karen M Knapp** in Shayne Chau, Emma Hyde, Karen Knapp, Christopher Hayre. Person-Centered Care in Radiology. Routledge Taylor and Francis. In print – released 2024

4. Victoria Stiles and **Karen Knapp**. Bone Health in Exercise Management for Referred Medical Condition. 1st Edition. 2022. Routledge. 9781315102399

5. Knapp Karen and Carver Elizabeth. Genitourinary Contrast Imaging. In Carver E, Carver B and **Knapp KM**. Carvers' Medical Imaging. Third Edition. Elsevier 2021 ISBN 978-0-7020-6955-0

6. **Knapp Karen**. Dual Energy X-ray Absorptiometry. In Carver E, Carver B and **Knapp KM**. Carvers' Medical Imaging. Third Edition. Elsevier 2021 ISBN 978-0-7020-6955-0

7. **Knapp K.** The Radiographer's Role in DXA: Sports medicine and MSK. In Imaging of Motion and Performance. Stress and Strain. Editors: Vieira A, Sconfienza L, Kainberger F . 1, 96-103. European Society of Radiology, ESR 29 Oct 2019.
8. S M Masudur Rahman Al Arif **Karen M Knapp** Greg Slabaugh. Shape-Aware Deep Convolutional Neural Network for Vertebrae Segmentation. In book: Computational Methods and Clinical Applications in Musculoskeletal Imaging. DOI: 10.1007/978-3-319-74113-0_2

9. S M Masudur Rahman Al Arif **Karen M Knapp** Greg Slabaugh. SPNet: Shape Prediction Using a Fully Convolutional Neural Network. September 2018. Lecture Notes in Computer Science. In book: Medical Image Computing and Computer Assisted Intervention – MICCAI 2018 DOI: 10.1007/978-3-030-00928-1_49

10. Mellor F, **Knapp KM**. Medical Imaging and Radiotherapy Research. In Medical Imaging and Radiotherapy Research. Editors: Ramlaul A. Churchill Livingstone Jun 2010 ISBN: 9780702031045

Papers in journals

1. S Rajan, C Eade, N Ashley, **K Knapp**, S Rowlands. Artificial intelligence for detecting subtle paediatric tibial fractures in children under three years: An analysis of YOLO architectures with convolutional Block attention. Radiography 2026;32 (5):103472

2. Helen Neilens, Victoria Allgar, Kayle-Anne Sands, Jade Chynoweth, Amber Lord, Paigan J Aspinall, Helen Hambly, Anton Barnett, Lindsay Skipper, Tom Bewick, Ian Maidment, Hairil Abdul Razak, Ruth Ashton, David Strain, **Karen Knapp**, Mark A Faghy. An open-label, clinical feasibility study of the efficacy of Remdesivir for Long-COVID. Pilot and Feasibility Studies. 2026/3/4.

3. J Hu, J Chen, S Wells, **K Knapp**. Development of a novel anthropomorphic bone phantoms for mimicking osteoporosis in medical imaging development. Bone 2026;205:117803

4. Lucy R Banfield, **Karen M Knapp**, Luke C Pilling, David Melzer, Janice L Atkins. Chondrocalcinosis and the haemochromatosis-linked HFE C282Y homozygous variant in the UK Biobank. EULAR Rheumatology Open. 2026;2(1):1-8

5. Fahad Alghamdi, Abasiama Dick Obotiba, Robert Meertens, Omar Alshalawi, Kinan Mokbel, William David Strain, **Karen M Knapp.** Assessment of Bone Mineral Density, Total Body Composition and Joint Integrity in Long COVID: A 12-Month Longitudinal Feasibility Study. Journal of Clinical Medicine. 2025;14(23)8558

6. Fahad Alghamdi, Robert Meertens, Abasiama Dick Obotiba, Lorna W Harries, Sarah Appleby, Kinan Mokbel**, Karen M Knapp**, William David Strain. Assessment of Health-Related Quality of Life and Biomarkers in Long COVID: A 12-Month Longitudinal Feasibility Cohort. Journal of Clinical Medicine. 2025;14(22)7931

7. L Rodrigues, A Breen, **KM Knapp**, JR Meakin. Comparing the number of landmark points used to create statistical shape models of the lumbar spine from MR images. Radiography 2025; 31 (5), 103009

8. JP McNulty, M Zanardo, C Buissink, R DeCoster, W Hennessy, K Knapp, B Kraus, L Lanca, S Lewis, B Mahlaola Tintswalo, M McEntee, D O'Leary, H Precht, T Starc, LA Rainford. Professional challenges faced by student radiographers during COVID-19. Radiography 2025; 31 (4), 103000

9. KJ Knight, M Beasley, J McConnell, T O'Regan, CM Alexander, T Donovan, H Probs, R Reeve, M Sharma, **KM Knapp**, HA McNair. Research culture, barriers and facilitators within the radiography workforce in the UK–results of a national survey. Radiography 2025; 31 (4), 102959

10. Ian Alberts, Stuart More, **Karen Knapp**, Riccardo Mei, Stefano Fanti, Clemens Mingels, Lorenzo Nardo, Nii Boye Hammond, Harish Nagaraj, Axel Rominger, Gary JR Cook, Don Wilson. Is Long–Axial-Field-of-View PET/CT Cost-Effective?

An International Health–Economic Analysis. Journal of Nuclear Medicine 2025; 66 (6), 954-960

11. A Algahtani, M Asiri, K Mokbel, R Meertens, J Fulford, WD Strain, **K Knapp**. Participant Perspectives on Pulse-Echo Ultrasound Technology vs. Dual Energy X-Ray Absorptiometry (DXA): A Comparative Study. In Vivo 2025; 39 (2), 909-916

12. Fahad Alghamdi, Kinan Mokbel, Robert Meertens, Abasiama Dick Obotiba, Mansour Alharbi, **Karen M Knapp**, William David Strain. Bone Mineral Density, Bone Biomarkers, and Joints in Acute, Post, and Long COVID-19: A Systematic Review. Viruses 2024, 16(11), 1694; https://doi.org/10.3390/v16111694.

13. M Sayed, **KM Knapp**, J Fulford, C Heales, SJ Alqahtani. The impact of X-ray scatter correction software on abdomen radiography in terms of image quality and radiation dose. Radiography 2024;30(4);1125-1135

14. KJ Knight, RJ Fordham, NJ Crabtree, **KM Knapp**. Evaluating the impact of COVID-19 on DXA waiting lists and osteoporosis prescription trends in England 2019–2023. Osteoporosis International, 2024;1-10

15. R Assiri, **K Knapp**, J Fulford, J Chen. Optimising total knee replacement imaging: a novel 3D printed PET/CT anthropomorphic phantom for metal artefact simulation. EJNMMI physics 2024;11(1):31

16. Banfield LR, **Knapp KM**, Pilling LC, Melzer D, Atkins JL. Haemochromatosis genetic variants and musculoskeletal outcomes: 11.5 year follow-up in the UK Biobank cohort study. 2023. JBMR Plus Volume 7, Issue 10, 1 October 2023, e10794

17. Malik H, Wood D, Stone O, Gough A, Taylor G, **Knapp K**, Heggs D, Appelboam A. Ultrasound Directed Reduction of Colles' Type distal radial fractures in ED (UDiReCT): a feasibility randomized controlled trial. 2023 Emergency Medicine Journal 40 (12), 832-839

18. Monteyne AJ, Coelho MOC, Murton AJ, Abdelrahman DR, Blackwell JR, Koscien CP, **Knapp KM**, Fulford J, Finnigan TJA, Dirks ML, Stephens FB, Wall BT. Vegan and Omnivorous High Protein Diets Support Comparable Daily Myofibrillar Protein Synthesis Rates and Skeletal Muscle Hypertrophy in Young Adults. The Journal of Nutrition. 2023;153(6)1680-1695.

19. D Smith, **K Knapp**, C Wright, R Hollick. Dual energy x-ray absorptiometry (DXA) extended femur scans to support opportunistic screening for incomplete atypical femoral fractures: A short term in-vivo precision study. Journal of Clinical Densitometry In Press Available online 8 December 2023;26(2):101352

20. M Sayed, **KM Knapp**, J Fulford, C Heales, S J Alqahtani. The principles and effectiveness of X-ray scatter correction software for diagnostic X-ray imaging: A scoping review. European Journal of Radiology 2022; 158:110600

21. **KM Knapp**, S Venner, JP McNulty, LA Rainford. The risk of burnout in academic radiographers during the COVID-19 pandemic. Radiography 2022;28(4):1010-1015

22. **KM Knapp**, S Venner, JP McNulty, LA Rainford. The challenges, coping mechanisms, and recovery from the initial waves of the COVID-19 pandemic among Academic Radiographers. Radiography 2022;28:S35-S40

23. Hu, J., M. Gundry, K. Zheng, J. Zhong, P. Hourigan, J. R. Meakin, C. P. Winlove, A. D. Toms, **K. M. Knapp** and J. Chen (2022). The biomechanics of metaphyseal cone augmentation in revision knee replacement. Journal of the Mechanical Behaviour of Biomedical Materials 131: 105233.

24. Assiri R, **Knapp K**, Fulford J, Chen J. Correlation of the quantitative methods for the measurement of bone uptake and plasma clearance of 18F-NaF using positron emission tomography. Systematic review and meta-analysis. Eur J Radiol 2022;146:110081

25. Malamateniou C, **Knapp KM**, Pergola M, Woznitzsa N, Hardy M. Artificial intelligence in radiography: Where are we now and what does the future hold? Radiography 2021;27:S58-S62

26. Meertens R, **Knapp KM**, Strain WD, Casanova F, Ball S, Fulford J, Thorn C. In vivo Measurement of Intraosseous Vascular Haemodynamic Markers in Human Bone Tissue Utilising Near Infrared Spectroscopy. Front Physiol 2021;12:78239

27. Hudson D, **Knapp K**, Benwell M. An evaluation of MRI lumbar spine scans within a community-based diagnostic setting. Musculoskeletal care. 2021;19(30:384-395

28. Rainford LA, Zanardo M, Buissink C, Decoster R, Hennessy W, **Knapp K**, Kraus B, Lanca L, Lewis S, Mahlaola TB, McEntee M, O'Leary D, Precht H, Starc T, McNulty JP. The impact of COVID-19 upon student radiographers and clinical training. Radiography 2020:27;464-474

29. Bleiker J, **Knapp KM**, Morgan-Trimmer S, Hopkins SJ. What medical imaging professionals talk about when they talk about compassion. J Medical Imaging and Radiation Sciences. 2020;51:S44-S52.

30. Reyes-Aldasoro C, Ngan KH, Ananda A, d'Avila Garcez A, Appelboam A, **Knapp K M**. Geometric semi-automatic analysis of radiographs of Colles' fractures. PlosOne 2020;15(9):e0238926

31. Luke J Connolly, Suzanne Scott, Carmelina M Morencos, Jonathan Fulford, Andrew M Jones, **Karen Knapp**, Peter Krustrup, Stephen J Bailey, Joanna L Bowtell. Impact of a novel home-based exercise intervention on health indicators in inactive premenopausal women: a 12-week randomised controlled trial. European Journal of Applied Physiology. 2020;4:771-782

32. McAnulla SJ, Ball SE, **Knapp KM**. Understanding student radiographer attrition: Risk factors and strategies. Radiography. 2020;26:198-204

33. Heales CJ, Summers IR, Fulford J, **Knapp KM**, Winlove CP. Investigation of changes in bone density and chemical composition associated with bone marrow oedema-type appearances in magnetic resonance images of the equine forelimb. BMC MUSCULOSKELETAL DISORDERS 2019;20:1-10

34. Rimes SJ, **Knapp KM**, Meertens RM, Fox DL. Computed tomography colonography: Radiographer independent preliminary clinical evaluation for intraluminal pathology. Radiography 2019;25:S4-S8

35. Meakin JR, Ames RM, Jeynes CCG, Welsman J, Gundry M, **Knapp K**, Everson R. The feasibility of using citizens to segment anatomy from medical images: Accuracy and motivation. PLoS ONE 201914(10): e0222523

36. KH Ngan, AA Garcez, **KM Knapp**, A Appelboam, CC Reyes-Aldasoro. Making densenet interpretable a case study in clinical radiology, 2019 medRxiv, 19013730

37. Alqahtani S, Wellbourn R, Meakin JR, Palfrey R, Rimes S, Thomson K, **Knapp K**. Increased radiation dose and projected radiation-related lifetime cancer risk in patients with obesity due to projection radiography. Journal of Radiological Protection 2019;39:38-53

38. Rachuba Sebastian, **Knapp Karen**, Ashton Lucy, Pitt Martin, Streamlining pathways for minor injuries in emergency departments through radiographer-led discharge. Operations Research for Health Care. 2018;19:44–56

39. Bleiker J, **Knapp KM**, Morgan-Trimmer S, Hopkins SJ. "It's what's behind the mask": Psychological diversity in compassionate patient care. Radiography 2018 24;;S28-S32

40. Meertens R, Casanova F, **Knapp KM**, Thorn C, Strain WD. Use of near-infrared systems for investigations of hemodynamics in human in vivo bone tissue: A systematic review.J Orthop Res. 2018;36(10):2595-2603

41. Gundry, M, **Knapp, KM**, Meertens RM, Meakin JR. Computer-aided detection in musculoskeletal projection radiography: A systematic review. Radiography 2018;24(2):165-174

42. Hopkins, S. J.; Toms, A. D.; Brown, M.; Appleboam, A.; **Knapp, K. M**. Disuse osteopenia following leg fracture in postmenopausal women: Implications for HIP fracture risk and fracture liaison services. Radiography 2018;24(2):151-158.

43. Al Arif SMMR, **Knapp KM**, Slabaugh G. Fully automated cervical vertebrae segmentation framework for x-ray images. Computer Methods and Programs in Biomedicine. 2018:157:95-111

44. Vlachopoulos D, Barker AR, Ubago-Guisado E, Ortega FB, Krustrup P, Metcalf B, Castro Pinero J, Ruiz JR, **Knapp KM**, Williams CA et al. The effect of 12-month participation in osteogenic and non-osteogenic sports on bone development in adolescent male athletes. The PRO-BONE study. Journal of Science and Medicine in Sport 2018;21(4):404-409

45. SM Al Arif, **Karen Knapp**, Greg Slabaugh. Shape-aware deep convolutional neural network for vertebrae segmentation. International Workshop on Computational Methods and Clinical Applications in Musculoskeletal Imaging. 2018:12-24

46. SM Masudur Rahman Al Arif, **Karen Knapp**, Greg Slabaugh. Spnet: Shape prediction using a fully convolutional neural network. Medical Image Computing and Computer Assisted Intervention–MICCAI 2018: Proceedings, Part I. 430-439.

47. Stiles VH, Metcalf B, **Knapp K**, Rowlands A. A small amount of precisely measured high intensity habitual physical activity predicts bone health in pre- and

post-menopausal women in UK Biobank. Int J Epidemiol. 2017 Dec 1;46(6):1847-1856

48. Vlachopoulos D, Barker AR, Ubago-Guisado E, Fatouros IG, **Knapp KM**, Williams CA, Gracia-Marco L. Longitudinal Adaptations of Bone Mass, Geometry and Metabolism in Adolescent Male Athletes. The PRO-BONE Study. Journal of Bone and Mineral Research. 2017;32:2269-2277

49. SMMR Al Arif, M Asad, M Gundry, **K Knapp**, G Slabaugh. Patch-based corner detection for cervical vertebrae in X-ray images. Signal Processing: Image Communication 59, 27-36

50. **Knapp KM**, Wright C, Clarke H, McAnulla SJ, Nightingale J. The academic radiography workforce: Age profile, succession planning and academic development. Radiography 2017;23:S48-S52

51. SMMR Al Arif, **K Knapp**, G Slabaugh. Region-aware deep localization framework for cervical vertebrae in X-ray images. Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: MICCAI 2017:74-82

52. Vlachopoulos D, Barker AR, Williams CA, Arngrímsson SA, **Knapp KM**, Metcalf BS, Fatouros IG, Moreno LA, Gracia-Marco L. The Impact of Sport Participation on Bone Mass and Geometry in Male Adolescents. Medicine and Science in Sports and Exercise 2017:**49**(2):317-326 01.

53. SM Masudur Rahman Al Arif, Michael Gundry, **Karen Knapp**, Greg Slabaugh. Global localization and orientation of the cervical spine in x-ray images. International workshop on computational methods and clinical applications for spine imaging. 2016:64-76

54. Hopkins SJ, Toms AD, Brown M, Welsman JR, Ukoumunne OC, **Knapp KM**. A study investigating short-and medium-term effects on function, bone mineral density and lean tissue mass post-total knee replacement in a caucasian female post-menopausal population: Implications for hip fracture risk. Osteoporosis International 2016;27(8):2567-2576

55. Bleiker J, **Knapp KM**, Hopkins S, Johnston G. Compassionate care in radiography, education and training: A post-Francis report review of the current literature and patient perspectives. Radiography Aug 2016 22;3:257-262

56. Bowtell JL, Jackman SR, Scott S, Connolly LJ, Mohr M, Ermidis G, Julian R, Yousefian F, Helge EW, Jørgensen NR, Fulford J, **Knapp KM**, Krustrup P. Short duration small sided football and to a lesser extent whole body vibration exercise induce acute changes in markers of bone turnover. Biomed Res Int. 2016;2016:3574258

57. SM Masudur Rahman Al Arif, Michael Gundry, **Karen Knapp**, Greg Slabaugh. Improving an active shape model with random classification forest for segmentation of cervical vertebrae. Computational Methods and Clinical Applications for Spine Imaging: 4th International Workshop and Challenge, CSI 2016, MICCAI 2016: 3-15

58. Dimitris Vlachopoulos, Alan R Barker, Craig A Williams, SIGURBJöRN A ARNGRíMSSON, **Karen M Knapp**, Brad S Metcalf, Ioannis G Fatouros, Luis A

Moreno, Luis Gracia-Marco. The impact of sport participation on bone mass and geometry in adolescent males. Med. Sci. Sports Exerc 2016;49:317-326

59. Dimitris Vlachopoulos, Alan R Barker, Craig A Williams, **Karen M Knapp**, Brad S Metcalf, Luis Gracia-Marco. Effect of a program of short bouts of exercise on bone health in adolescents involved in different sports: the PRO-BONE study protocol. 2015;15:1-10

60. Meertens RM, **Knapp KM**, Strain WD, Casanova F. The Effects of Lumbar Sympathectomy on Bone and Soft Tissue Haemodynamics of the Leg recorded using Near Infrared Spectroscopy: A Case Report. Journal of Biomedical Engineering and Informatics 2016:3:28-32

61. Rimes SJ, Fox D, **Knapp KM**, Meertens R. The development and evaluation of an audit tool for measuring reporting accuracy of radiographers compared with radiologists for intra-luminal pathology detected at computed tomography colonography (CTC). 2015 Radiography 21(3):264-268

62. Birch C, **Knapp K**, Hopkins S, Gallimore S, Rock B. SpineAnalyzer™ is an accurate and precise method of vertebral fracture detection and classification on dual-energy lateral vertebral assessment scans. 2015 Radiography 21(3):278-281

63. **Knapp K M**, Hopkins, S J, Welsman J R, Fogelman I, Blake G M. Obesity increases precision errors in total body dual x-ray absorptiometry measurements. Journal Clinical Densitometry. 2015;18:209-216

64. Narang B, Phillips M, **Knapp K**, Appelboam A, Reuben A, Slabaugh G. Semi-Automatic Delineation of the Spino-Laminar Junction Curve on Lateral X-ray Radiographs of the Cervical Spine. MEDICAL IMAGING 2015: IMAGE PROCESSING. 9413:729-734

65. B Jebri, M Phillips, K Knapp, A Appelboam, A Reuben, G Slabaugh. Detection of degenerative change in lateral projection cervical spine x-ray images. Medical Imaging 2015: Computer-Aided Diagnosis 9414, 18-25

66. Connolly LJ, Scott S, Mohr M, Ermidis G, Julian R, Bangsbo J, Jackman SR, Bowtell JL, Davies RC, Hopkins SJ, Seymour, R, **Knapp KM**, Krustrup P, Fulford J. Effects of small-volume soccer and vibration training on body composition, aerobic fitness, and muscular PCr kinetics for inactive women aged 20-45 Journal of Sport and Health Science 2014;3:284-292

67. Cantin P, **Knapp KM**. Diagnostic image quality in gynaecological ultrasound. Who should measure it, what should we measure and how? Ultrasound. Ultrasound 2014:22:44-51

68. Hopkins S J, Welsman J R, **Knapp K M**. Short-term precision error in dual energy x-ray absorptiometry, bone mineral density and trabecular bone score measurements; and effects of obesity on precision error. J Biomedical Graphics and Computing. 2014;4(2):8-14.

69. Meakin JR, Welsman, JR, Fulford J, Seymour R, **Knapp KM**. The relationship between sagittal curvature and extensor muscle volume in the lumbar spine. Journal of Anatomy. 2013;222:608-14

70. Hopkins S, Smith C, Toms A, Brown M, Welsman J, **Knapp K**. Evaluation of a dual-scales method to measure weight-bearing through the legs, and effects of weight-bearing inequalities on hip bone mineral density and leg lean tissue mass. Journal of Rehabilitation Medicine. 2013;45(2):206-10.

71. Griffin JGL, **Knapp KM**, Pearce G. Static and Mobile DXA scanner in–vivo cross-calibration study. Radiography 2013;19:7-10

72. Wood G, **Knapp KM**, Rock B, Cousens C, Roobottom C, Wilson MR. Visial expertise in detecting and diagnosing skeletal fractures. Skeletal Radiology. 2013 Feb;42(2):165-72.

73. **Knapp KM,** Welsman JR, Hopkins SH, Fogelman I, Blake GM. Obesity increases precision errors in dual x-ray absorptiometry measurements. J Clin Densitometry. 2012:15:315-9

74. Allison R, **Knapp KM.** Spasticity management with botulinum toxin: development and evaluation of a tool for audit. J Rehabilitation Medicine 2012:44(7):558-61

75. Bleiker J, **Knapp KM**, Frampton I. Teaching patient care to students: A blended learning approach in radiography education. Radiography. 2011;17:235-240

76. Hayes S, Miller J, **Knapp K**. Is there a correlation of bone mineral density measurement in premenopausal women and their socio-demographic profile? Clinical Chiropractic 12/2012; 15(s 3–4):178.

77. **Knapp KM**, Rowlands AV, Welsman JR and MacLeod KM. Prolonged unilateral disuse osteopenia 14 years post external fixator removal: A case history and critical review. Case Reports in Medicine. 2010:629020

78. Moger CJ, Barrett R, Bleuet P, Bradley DA, Ellis RE, Green EM, **Knapp KM**, Muthuvelu P, Winlove CP. Regional variations of collagen orientation in normal and diseased articular cartilage and subchondral bone determined using small angle X-ray scattering (SAXS). Osteoarthritis Cartilage. 2007 Jun;15(6):682-7.

79. J Mansfield, CP Winlove, JJ Moger, **K Knapp**, SJ Matcher. Second-harmonic and two-photon imaging and polarimetry of articular cartilage. Multiphoton Microscopy in the Biomedical Sciences VII 6442, 299-309

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90. **Knapp K M**, Blake G M, Fogelman I, Doyle D V & Spector T D. *Multi-site Quantitative Ultrasound: Colles' Fracture Discrimination on Postmenopausal Women.* Osteoporosis Int 2002 13:474-479

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Review articles in journals

1. Hayre CM, **Knapp KM**. Leading radiography research to deliver clinical outcomes. Journal of Medical Imaging and Radiation Sciences 2022;53(4):S35-S37

2. F Alghamdi, Owen R, Ashton REM, Obotiba AD, Meertens, RM, Hyde E, Faghy MA, **Knapp KM**, Rogers P, Strain WD. Post-acute COVIC syndrome (long COVID): What should radiographers know and the potential impact for imaging services. Radiography 2022;28:S93-S99

3. SJ Alqahtani, **KM Knapp**. Imaging patients with obesity. Journal of Medical Radiation Sciences 2022;69(1):3-4

4. **Knapp KM,** White H, McPake M, McConnell J, Probst H, Samuels L, Nolan R, Malamateniou C, Sloane C, Vallis J, Beardmore C. The new College of Radiographers' Education and Career Framework (ECF) for Radiography. Motivations, Process and Aspirations for the Future: A Discussion to Inform the ECF Development. Imaging and Oncology. 2021; 38-46

5. **KM Knapp**, N Courtier. The future role of radiographers. Radiography 2021; 27:S1-S2

6. Bleiker J, Morgan-Trimmer S, **Knapp K M**, Hopkins S. Navigating the maze: Qualitative research methodologies and their philosophical foundations. Radiography 2019;25:S4-S8 Editors' choice 2019

7. J Arnold, P Ewings, L Handel, M Langdon, H Powell, Z Rhydderch-Evans, W Stone, S Woodland, L Hoade, J Bleiker, **KM Knapp**. Fragile: Please handle with care. 2018;24:S9-S10

8. **Karen M Knapp**, RM Meertens, Richard Seymour. Imaging and opportunistic identification of fractures. Geriatric Medicine 2018;48:10-12

9. Gundry M, Hopkins S, **Knapp K**. A Review on Bone Mineral Density Loss in Total Knee Replacements Leading to Increased Fracture Risk. 2017 Clinical Reviews in Bone and Mineral Metabolism 1-13 27 Oct 2017

10. JP McNulty, **KM Knapp**, P Brown. Radiography education in the spotlight. Radiography 2017;23:S1-S2.

11. Meertens RM, Strain WD, **Knapp KM**. A Review of the Mechanisms, Diagnosis and Preventative Treatment of Osteoporotic Fragility Fractures in Patients With Type 2 Diabetes Mellitus. Journal of Endocrinology and Metabolism 2015; 5:157-162

12. **Knapp KM,** Meertens RM, Ashton LR, Hopkins SJ. Dual energy x-ray absorptiometry beyond bone mineral density measurement: vertebral fracture assessment. Imaging and Oncology. 2015; 48-53

13. **Knapp KM**, Griffin JGL, Blake GM. Dual Energy X-ray Absorptiometry: Quality assurance and governance. Osteoporosis Review. 2014. Vol 22;No1:1-6

14. **Knapp KM**, Griffin J, Peel NFA. Dual Energy X-ray Absorptiometry: Artifacts and Pitfalls in Interpretation. Osteoporosis Review . 2014. Vol 22;No1:10-16

15. **Knapp KM**. Dual energy absorptiometry sports performance assessment. Osteoporosis Review 2014; 22:26-27 (Informed DH Consultation)

16. **Knapp KM**, Meertens R, Seymour R. Imaging in osteoporosis diagnosis and management. Geriatric Medicine. 2013:10 http://www.gmjournal.co.uk/imaging_in_osteoporosis_23622320680.aspx

17. **Knapp KM** and Green S. The Reporting of Dual Energy X-ray Absorptiometry Scans by Radiographers. Osteoporosis Review 2009;17(1):22-25

18. **Knapp KM**. Quantitative ultrasound and bone health. Salud Publica Mex. 2009;51 Suppl 1:S18-24

Research Impact and Knowledge Exchange

Official Reports

- All Party Parliamentary Group report on Osteoporosis 2023. Led by Royal Osteoporosis Society. *This outcomes from this are being used to influence*

Government and fracture liaison services have already been agreed for Scotland and Wales.

- NICE panel member for osteoporosis clinical guidelines (update) 2023-2026
  The new NICE guidelines will be the evidence-based and cost-effectiveness guidelines for the management and treatment of osteoporosis across the UK. These have not been updated for 15 years, so are eagerly awaited by the community.

Working Papers

- College of Radiographers Education Career Framework 2022 (Chair of the committee)
- Skills for Health National Occupational Standard to Produce DXA scans for diagnostic purposes Dec 2018
- Role of the Radiographer in DXA and osteoporosis services 2017-2018 – College of Radiographers
- Department of Health Justification working party regarding DXA Sports Performance Assessment 2014
- College of Radiographers Framework for Learning and Teaching 2005

Education

Courses / Modules Taught
Module Lead*:

Undergraduate

- *Clinical placement 1 (PAM1007) 2005-06 (and curriculum development)
- *Psychology (PAM1008) 2007-08
- *Pathology for radiographers PAM2002 (developed in 2005 and taught until 2014 PAM2012)
- *Skeletal Image interpretation (PAM3006) 2007-08 until 2013-14
- *Project studies (PAM2005) 2009-10
- PAM3012 2018-2021
- PAM3004 2009-2022
- PAM1020 2020-2021
- PAM2006 2020-2021

Postgraduate

- *Musculoskeletal pathology 2016-17-2017-18
- *Advanced clinical research 2009-10 to 2014-15
- *Small scale service related project 2009-10 to 2014-15
- HPDM156 2023-24
- HPDM065 2023-24

New programmes
I developed the following new programmes:

- BSc (Hons) Medical Imaging (Diagnostic Radiography) programme
- Doctorate of Clinical Research (DClinRes)
- MbyRes in Medical Imaging

Mentored Development

- BSc (Hons) Diagnostic Radiography and Imaging Degree Apprenticeship
- MSc Skeletal Image Interpretation

Grants for Educational Purposes

1. 2022-2025 **Health Education England** PI Simulation training and placements expansion fund. £222,147 (External)

2. 2021-2022 **Health Education England** Co-I Simulation equipment training fund. Application for simulation resources to enhance undergraduate inter-professional education to improve patient care and student preparedness for practice £250,000 (External)

3. 2020 **Health Education England** PI Support for diagnostic radiography student retention and clinical placement expansion. £27,779 (External)

4. 2014-2015 **Health Education South West** (HESW) PI Demonstrator for Advanced (Radiographer) Practice in Emergency Care (DARP) Sep 2014-Sep 2015 £57,000 (External)

5. 2010-2011 **Higher Education Academy**. Develop blended-learning psychology teaching materials for undergraduate healthcare students. June 2010 – June 2011 £6,000

External Examining

Undergraduate and postgraduate Taught Courses

1. External examiner for Canterbury Christ Church University (Undergraduate programme) 2007-2010
2. External examiner for Kingston and St Georges (postgraduate programmes) 2011-2014
3. External moderator for postgraduate research module, Unitec Institute of Technology, Aukland, New Zealand. 2010
4. External examiner for City University, London 2013- 2017
5. External examiner for L_Universitá ta' Malta 2023-2027

PhD Examinations

1. External PhD examiner for the Department of Radiography, City University, London
2. External PhD examiner for the Department of Radiography, City University, London
3. External PhD examiner for Swansea University
4. External Prof Doc examiner for Glasgow Caledonian University, 2015
5. External PhD examiner for University College Dublin 2017, 2022
6. External PhD examiner for the University of Sheffield 2019
7. External PhD examiner for Imperial College, 2019
8. External PhD examiner for Glasgow Caledonian University, 2020
9. External PhD examiner for University of Salford 2021 & 2023
10. External PhD examiner for University of Johannesburg 2022
11. External PhD examiner for Ulster University 2023
12. External PhD examiner for University of Sydney 2024

Internal PGR examiner

1. 1x DClinRes

2. 2 x PhD

6. Postgraduate Research

MbyRes

5 Completed students

DClinRes

3 Completed students

Completed PhD Students

17 Completed students

Current PhD students

10 Current students

Leadership and management

Faculty or discipline management and administrative responsibilities

- Director of Education for Medical Imaging 2010-2013
- Head of Medical Imaging 2013-2022
- Head of Health and Care Professions 2022-current

University appointments, membership of Committees, Working Parties

- Senate representative on Council 2023-July 2026
- Senate 2022-current
- Senate 2014-2020
- 2014-2015 Professional Doctorates working group to harmonise the professional doctorate regulations and publish in the TQA manual.
- University of Exeter Workload task and finish group 2021

External recognition

General Editorship of Journals / Series
Guest Editor for the following special issues:

The Future Role of the Radiographer. Edited by **Karen Knapp**, Nick Courtier. Volume 27, Supplement 1, S1-S88 (October 2021)

Radiography Education. Edited by Jonathan McNulty, Paul Brown, **Karen Knapp.** Volume 23, Supplement 1, S1-S76 (September 2017)

Honours and Awards

- Young Investigator Award at the Seventh Bath Conference on Osteoporosis, April 2000.

- Young Investigator Award at the Ninth Bath Conference on Osteoporosis, June 2003.
- Professions Allied to Medicine award at the Tenth National Osteoporosis Society Conference, December 2004
- Allied Health Professions award at the Eleventh National Osteoporosis Society Conference, June 2006
- Supervised the project of Chaleigh Franklin and Katie Hart who won the Forder Memorial prize for best student paper proffered at the UKRC 2007
- Supervised the project of Joanna Wilson who won the Forder Memorial prize for best student paper proffered at the UKRC 2008
- University of Exeter Teaching Fellowship, Jan 2013
- Runner up in the University of Exeter Teaching awards for Research Inspired Teaching May 2013
- Supervised the project of Victoria Charles who won the Forder Memorial prize for the best student paper proffered at the UKRC June 2013
- Runner up in the University of Exeter Teaching awards for Research Inspired Teaching May 2014
- Co-author, best scientific research poster at the UKRC June 2017
- Co-author, Editors' choice paper in Radiography 2019
- Co-author Editor's choice paper in JMIRS 2020
- Co-author commended poster ROS conference 2023 x 2

Membership and official positions in any professional bodies

- 2023-2025 NICE Clinical Guidelines Committee for Osteoporosis Guidelines (update)
- 2021-2023 European Federation of Radiographer Societies (EFRS) Research Committee Chair
- 2021-ongoing European Federation of Radiographer Societies (EFRS) Research Committee member and immediate past chair
- 2022-2030 Royal Osteoporosis Society Conference organising committee
- 2021-2024 European Society of Radiology, ECR conference organising committee
- 2021- ongoing NIHR PCAF Panel member
- 2020-2022 Society and College of Radiographers – Chair for Education and Career Framework (version 4)
- 2020-ongoing Royal Osteoporosis Society Bone Densitometry Training and Accreditation panel chair and Clinical and Research Committee member
- 2014-2020 Radiography Editorial board
- 2006-2015 College of Radiographers research group member
- 2006-2020 Royal Osteoporosis Society Bone Densitometry Training and Accreditation panel